

What is a GPU and TPU?

What is a GPU?

A GPU (Graphics Processing Unit) is a specialized computer chip designed to handle hundreds of tasks at the exact same time (known as parallel processing).

While a standard computer processor (CPU) acts like a single brilliant mathematician solving one hard problem at a time, a GPU is like a stadium filled with thousands of math students solving thousands of simple problems all at once.

Where is a GPU Used?

- **3D Gaming & Graphics:** Powers visual effects, shadows, and realistic textures in modern video games and 3D animation.
- **Video Editing & Rendering:** Speeds up high-definition (4K/8K) video editing, visual effects (VFX), and 3D rendering in tools like Premiere Pro or Blender.
- **Artificial Intelligence & Machine Learning:** Trains and runs AI models (like image generators, speech tools, and large language models) because AI heavy-lifting relies on parallel math.
- **Scientific Simulations & Mining:** Used for weather forecasting, medical research simulations, and crypto mining.

What is a TPU?

A TPU (Tensor Processing Unit) is a custom-designed hardware chip built by Google specifically to accelerate Artificial Intelligence and Machine Learning.

Unlike a GPU, which was originally built for rendering graphics and later adapted for AI, a TPU was built from scratch for AI math (specifically matrix and tensor operations). It sacrifices versatility to achieve extreme speed, low latency, and energy efficiency for AI workloads.

Where is a TPU Used?

- Google Services: Powers many of Google's everyday tools, including Google Search, Google Translate, Google Photos (for visual recognition), and YouTube recommendations.
- Training Large Language Models: Used inside Google Cloud to train and run massive generative AI models like Gemini.
- Enterprise AI on Google Cloud: Used by companies renting compute power on Google Cloud to run high-volume Machine Learning systems efficiently.

Quick Comparison

Feature	GPU (Graphics Processing Unit)	TPU (Tensor Processing Unit)
Origin	Created for 3D graphics & video games	Created specifically for AI & Machine Learning
Flexibility	Highly flexible (games, rendering, AI, science)	Specialized (optimized purely for AI tensor math)
Availability	Available everywhere (NVIDIA, AMD, Intel)	Primarily available via Google Cloud Platform
Best For	Gaming, general AI development, rendering	Heavy, large-scale AI training & serving at scale